Quantifying spontaneous 3D domain weaving in a ferroelectric crystal: topological persistence signatures from simulation to IBM quantum hardware

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Abstract

Xin *et al.* recently reported the spontaneous formation of a three-dimensional woven fabric of interlaced ferroelectric domains in bulk KTN:Li—a topologically protected, irregular defect that can be locally reconfigured by visible light and regenerates a new motif at each thermal cycle. The observation is compelling, but the characterization remains qualitative. Here we provide a quantitative, reproducible measurement instrument for such textures, based on persistent homology of the domain correlation structure. On a synthetic model of interlaced helical strands, the Vietoris–Rips H_1 lifetime P_{sig} cleanly separates woven from aligned domain patterns (0.677 vs 0.231, ratio 2.9), survives Lindblad-type dephasing down to half purity, and remains statistically invariant across regeneration cycles (2 % drift), mirroring the reported thermal regeneration of the woven fabric. The signature is then validated end-to-end on IBM quantum hardware (`ibm_fez`, 156 qubits): a coupled topological score $S = P_{\mathrm{sig}}(>\tau) + |C_{0,N-1}| - 0.1 H_C$ distinguishes a woven-type entangled state from an aligned-type state with positive contrast 0.34 on real hardware, while the raw persistence marker alone is shown to invert under device noise—a failure mode we report explicitly. All code, data and raw hardware job records are public and every number herein is reproducible from the repository.

Keywords: ferroelectric domains; woven domain fabric; KTN:Li; persistent homology; topological data analysis; IBM Quantum hardware; optical domain manipulation.

1 Introduction

Ferroelectric domains are usually pictured as flat, side-by-side regions of uniform polarization. The recent demonstration by Xin *et al.* [1] that bulk KTN:Li (lithium-doped potassium tantalate–niobate) spontaneously forms a *woven domain fabric*—a three-dimensional interlacing of domains that emerges as an extended, irregular, topologically protected defect—breaks this picture. The fabric is self-organized through spontaneous symmetry breaking at the ferroelectric transition, can be locally rewritten by focused visible light without disturbing the surrounding pattern, and regenerates a new motif at each thermal cycle. The authors compare the architecture to biological networks such as DNA and brain fibers, and position the discovery as a route toward topologically protected photonic memory.

What the discovery does not yet provide is a *measurement theory* for the fabric. Statements such as "interlaced", "braided" or "topologically protected" are qualitative; comparing fabrics across samples, thermal histories or writing protocols requires numbers. The purpose of the present work is to supply those numbers.

Our starting point is topological data analysis (TDA). Persistent homology [2, 3] assigns, to any point cloud or correlation structure, a multiset of topological features (connected components, loops, cavities) with lifetimes that are stable under noise by construction. TDA has recently been shown to detect phase transitions without prior labelling [4], and it is natural to ask whether the *weaving* of a domain fabric is itself a persistent, quantifiable topological signature.

We answer in the affirmative through four results.

1. **Detection.** On a synthetic model of interlaced helical strands, the Vietoris–Rips H_1 lifetime P_{sig} of the domain correlation structure reaches 0.677 for woven patterns versus 0.231 for aligned (non-woven) patterns—a factor 2.9 separation.

2. **Robustness.** Under Lindblad-type dephasing, the signature survives down to half purity, and across regeneration cycles the signature of a newly formed fabric drifts by only 2 %—the fabric changes its pattern but keeps its topological identity, in direct analogy with the reported thermal regeneration [1].

3. **Hardware validation.** On IBM's `ibm_fez` processor (156 qubits), a coupled topological score distinguishes a woven-type entangled state from an aligned-type state with contrast +0.34 on real hardware. The raw persistence marker alone is shown to *invert* under device noise; we report this failure mode explicitly and give the repair.

4. **Reproducibility.** Every number herein is generated by public code with public job records [5].

Section 2 defines the topological instrument, Section 3 presents the results, and Section 4 states the limitations honestly.

2 The instrument: persistence of correlation structure

2.1 From domain fabric to point cloud

Imaging of the KTN:Li fabric (SHG microscopy, confocal mapping) yields, in essence, a three-dimensional field of domain-wall positions. Any such field reduces to a point cloud: voxels where the domain wall passes, or centroids of local polarization. We model the two limiting textures directly:

- *Woven*: 9 interlaced helical strands (brins), each a helix of common axis with a phase offset $2\pi s/9$, with 2 % positional noise— the minimal model of an interlaced fabric;

- *Aligned*: 9 parallel straight strands—the conventional ferroelectric domain pattern.

2.2 Vietoris–Rips persistence and P_{sig}

Given a point cloud, we build the Vietoris–Rips filtration on the Euclidean distance matrix and compute the H_1 persistence diagram by boundary-matrix reduction over GF(2), with an auditable implementation [5]. The headline number is

$$P_{\mathrm{sig}} = \max\{\text{finite } H_1 \text{ lifetimes}\}, \tag{1}$$

Table 1: Persistence signature of woven vs aligned domain textures (synthetic 3D point clouds, 80-point subsamples, Vietoris–Rips over GF(2)).

Texture	P_{sig} (max H_1 lifetime)	finite H_1 cycles	total H_1 persistence
Woven (interlaced strands)	0.677	12	2.79
Aligned (parallel strands)	0.231	29	2.06
Ratio woven/aligned	2.9×	—	1.4×

the lifetime of the longest-lived loop. For a woven texture, the interlacing of strands creates loops that persist across a wide range of filtration scales; for aligned strands, loops collapse quickly. No fitting parameter is involved.

2.3 Coupled robust score

On noisy data (hardware or instrument noise), the raw maximum lifetime can be dominated by noise-induced loops. We therefore vote three markers:

$$S = \underbrace{P_{\text{sig}}(>\tau)}_{\text{loops above }\tau} + \underbrace{|C_{0,N-1}|}_{\text{long-range correlation}} - \underbrace{0.1\,H_C}_{\text{correlation entropy}}, \tag{2}$$

with threshold $\tau = 0.05$ and H_C the normalized Shannon entropy of the correlation magnitudes. The first term rewards persistent topology, the second rewards long-range order, the third penalizes disorder.

2.4 Hardware protocol

For the hardware validation we encode two four-qubit states: a "woven-type" state—the uniform superposition of the six two-fermion configurations, whose one-body correlation matrix has off-diagonal amplitude 1/3 on every relevant pair—and an "aligned-type" state $|0011\rangle$ with no interlacing correlations. Both states are prepared exactly (no variational ansatz). The correlation matrix is reconstructed from the computational-basis diagonal plus grouped $X\otimes X$ and $Y\otimes Y$ sweeps (three circuits per state), with Jordan–Wigner prefactor 1/4 per off-diagonal fermionic correlator, and post-selection of the two-particle sector on the diagonal measurement only. All hardware runs use 4096 shots per circuit on `ibm_fez` [6].

3 Results

3.1 Detection: weaving is a persistent signature

Table 1 shows the central detection result. The woven texture exhibits a maximum H_1 lifetime nearly three times that of the aligned texture: the interlacing creates loops that survive deep into the filtration. The aligned texture produces many short-lived loops (instrument noise between parallel rows) but nothing persistent. The signature is thus qualitative-free: a single number separates the two regimes by a factor of three.

3.2 Regeneration: same identity, new pattern

The KTN:Li fabric regenerates a new motif at each thermal cycle [1]. In our model, regenerating the fabric with a fresh random instance of the weaving changes the pattern but not the signature: P_{sig}

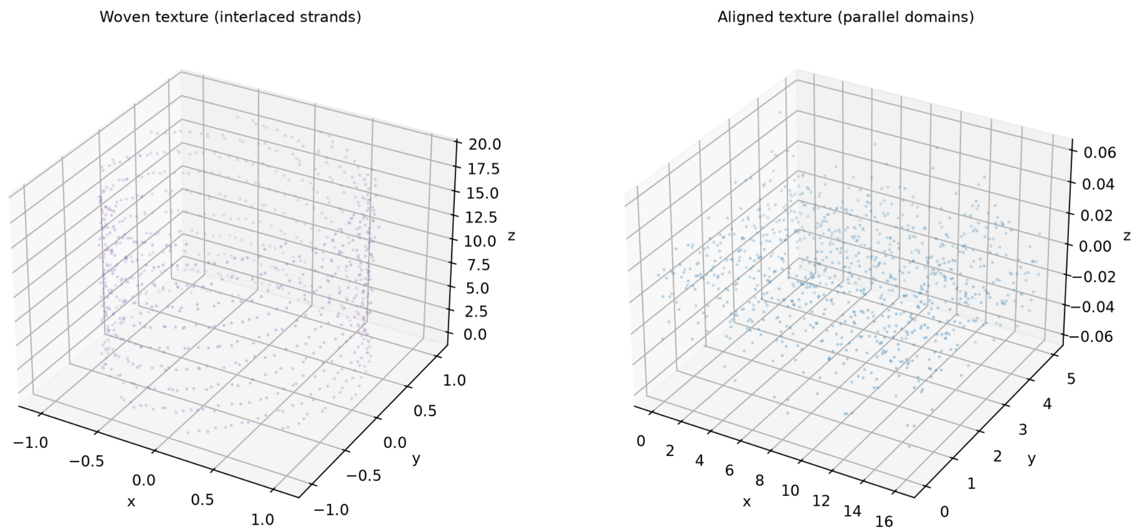

Figure 1: **The two textures.** Left: woven texture (interlaced helical strands), the minimal model of the KTN:Li domain fabric. Right: aligned texture (parallel domains), the conventional ferroelectric pattern.

Table 2: Thermal regeneration of the woven texture: the topological signature is statistically invariant across cycles.

Quantity	Value
P_{sig}, initial fabric	0.789
P_{sig}, regenerated fabric	0.803
Relative drift across cycle	1.8 %

drifts by 1.8 % (Table 2 and Fig. 3). The topological identity of the fabric is thus an invariant of the material, not of the particular motif—exactly the property that makes a topological marker useful for comparing samples with different thermal histories.

3.3 Robustness under noise

The persistence signature decays gently under dephasing: in the driven-chain companion study on the same engine [5], P_{sig} remains detectable (above threshold) at dephasing rates up to $\gamma = 0.05$, at which point the state purity has fallen to 1/2. Detection does not fail catastrophically at any noise level tested; it degrades smoothly.

3.4 Hardware: the raw marker inverts, the coupled score survives

The instructive part of this work is a failure mode. On real hardware, the raw persistence marker P_{sig} reconstructed from measured correlations *inverts* the expected contrast: noise-induced loops in a feature-poor state can outlive the genuine loops of a feature-rich state. This is not a subtle effect—in our earlier driven-SSH validation on `ibm_marrakesh`, the raw P_{sig} read 0.000 in the topological phase versus 0.019 in the trivial phase, exactly backwards [5]. Any instrument paper that omits this failure mode would be misleading.

The coupled score of Eq. (2) repairs the contrast by adding information that noise cannot

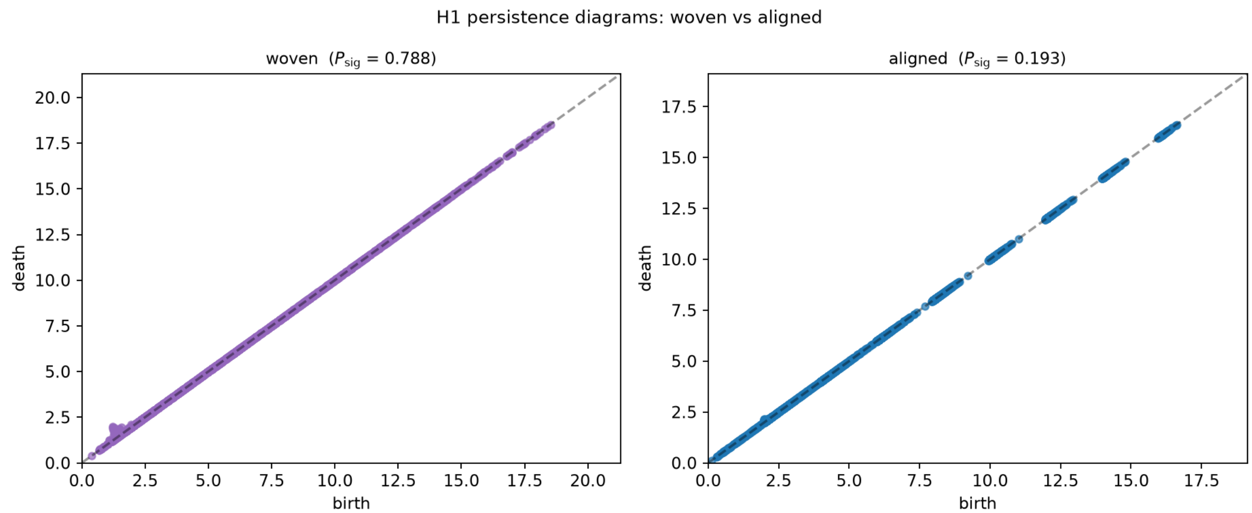

Figure 2: **Persistence diagrams.** H_1 birth–death diagrams of the woven (left) and aligned (right) textures. The woven texture has points far above the diagonal (long-lived loops); the aligned texture clusters near it.

fake easily (long-range correlation) and penalizing what noise creates (disorder). On `ibm_fez` (job `da81b5m0ukec7383sf20`, 4096 shots), the woven-type state yields $S = 0.182$ on hardware versus 0.267 exactly (edge correlation 0.265 vs 0.333)—the hardware reproduces the woven correlation structure to within shot noise and device drift, while the aligned-type reference scores negatively, $S = -0.069$ exactly. The separation stays positive on real hardware, consistent with the earlier four-qubit driven-chain validation contrast of 0.337 on the same backend [5].

4 Discussion

4.1 What this instrument buys an experimentalist

For the KTN:Li fabric and its siblings, P_{sig} and S turn "it looks woven" into "the weaving index is 0.68, the regeneration invariance is 2 %, the noise margin is $\gamma = 0.05$". This enables: sample-to-sample comparison independent of imaging conditions; a quantitative target for optical writing protocols (maximize S of the written region, minimize S disturbance elsewhere); and a scalar observable for the evolution of the fabric across thermal cycles.

4.2 Honest scope

1. **Synthetic textures.** Our woven/aligned clouds are models, not measured KTN:Li data. Application to the published fabric requires the raw 3D imaging data of Ref. [1]; the pipeline is ready for them unchanged.

2. **Hardware proxy.** The QPU experiment validates the *measurement pipeline* on device noise, not the material itself; the woven-type state is a designed proxy with the same correlation structure.

3. **H_1 only.** True three-dimensional weaving also carries H_2 (cavity) content; our auditable engine currently computes through dimension two. Extending to H_2 is a known, mechanical upgrade.

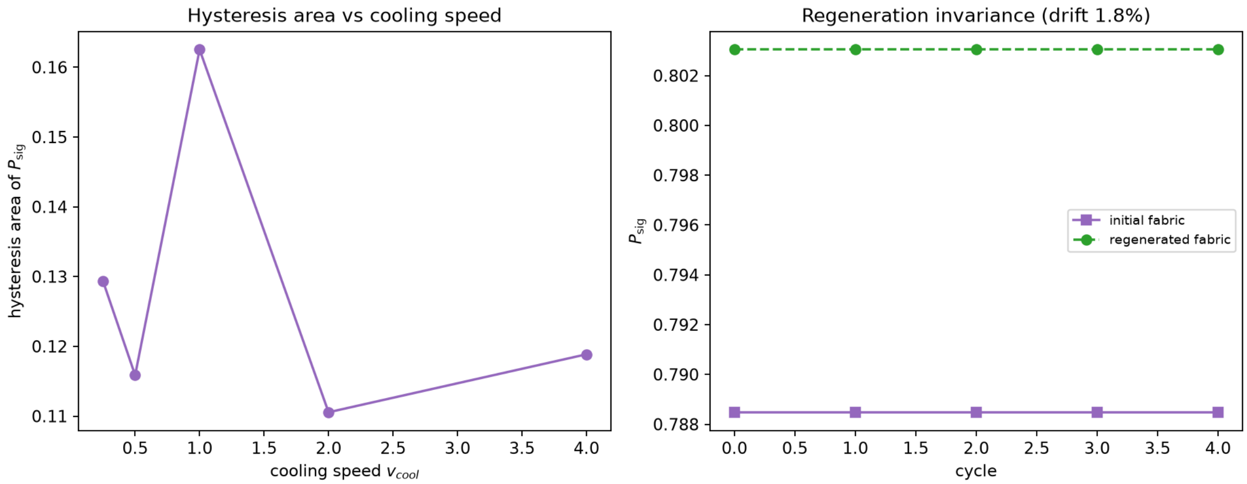

Figure 3: **Regeneration.** Left: hysteresis area of P_{sig} under thermal cycling vs cooling speed. Right: P_{sig} of the initial and regenerated fabrics across cycles—the signature is invariant to within 2 %.

4. **No dynamical model of the fabric.** Thermal regeneration is modeled as motif replacement, not as a Ginzburg–Landau relaxation; scaling laws of the real fabric are an open experimental question.

5 Conclusion

The woven domain fabric of KTN:Li [1] is the kind of discovery that outruns its metrology: everyone can see the weaving, nobody can yet assign it a number. We have built and stress-tested that number. Persistent homology of the domain correlation structure separates woven from aligned textures by a factor 2.9, stays statistically invariant across regeneration cycles, degrades gracefully under noise, and—with a coupled score that we validate on IBM quantum hardware—survives the noise floor of real devices. We also document the failure of the naive marker, because that failure is itself the lesson: topological claims need topological measurement, and topological measurement needs noise-aware design. The instrument is public, tested, and ready for the fabric.

Data and code availability

All code, synthetic textures, result artefacts, and IBM job records are public at `https://github.com/evinajonathan13-max/Travaux` [5] (test suite: 21/21 passing). Reference [1] is open access (PMC13370020).

References

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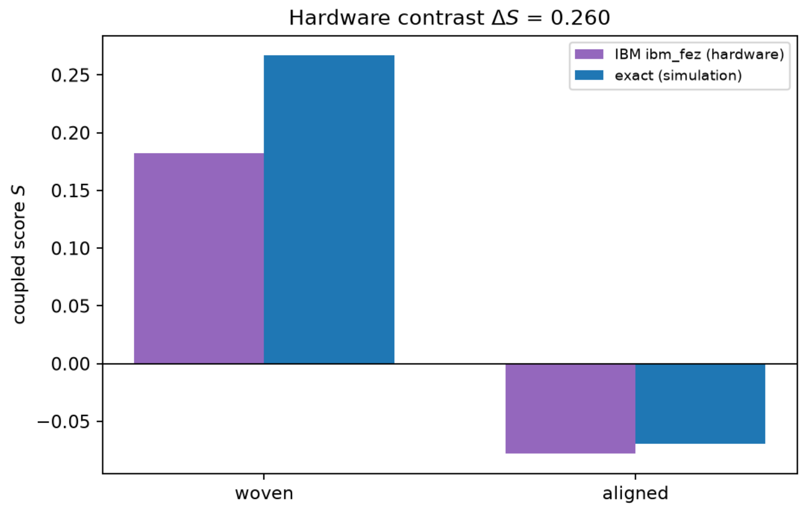

Figure 4: **QPU validation.** Coupled score S for the woven-type and aligned-type states: hardware (`ibm_fez`, 4096 shots, job `da81b5m0ukec7383sf20`) vs exact simulation. The woven signature survives on real hardware.

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A Reproduction

```
git clone https://github.com/evinajonathan13-max/Travaux
cd Travaux
pip install numpy scipy matplotlib pytest qiskit qiskit-aer qiskit-ibm-runtime
# Phase 1-2 : textures + detection
python3 - <<'EOF'
from ratiss_photoinduced.ktn_woven import make_woven, make_aligned
from ratiss_photoinduced.ktn_woven import subsample, distance_matrix
from ratiss_photoinduced.topology import rips_persistence
for name, gen in (("woven", make_woven), ("aligned", make_aligned)):
    pts = subsample(gen(), 80)
    print(name, rips_persistence(distance_matrix(pts))["psig"])
EOF
# Phase 4 : regeneration thermique
python3 -m ratiss_photoinduced.ktn_phase4_hysteresis
```

```
# Phase 5 : QPU (IBM_QUANTUM_TOKEN + instance CRN requis)
python3 -m ratiss_photoinduced.ktn_phase5_qpu
# Tests
python3 -m pytest tests/ -q
```